METRIC

MILITARY SPECIFICATION

BATTERY CELLS, STORAGE, SILVER-ZINC ALKALINE TYPE (FOR NR-1) (METRIC)

This amendment forms a part of Military Specification DOD-B-24507B(SH), dated 18 January 1978.

PAGE 4

3.4.6.5(c), line 1: Delete and substitute: "The positive absorbent material shall be 0.08 to 0.17 mm thick and the negative absorbent material shall be 0.12 to 0.17 mm thick."

3.4.6.5(d): Delete and substitute:

"(d) The main separator shall be Dupont Cellophane PUDO-193 (battery grade) as is or treated alone or in combination with Union Carbide Fibrous Sausage Casing (battery grade), or equal. The overall wet thickness of the combined layers of this material when swollen in the specified electrolyte, shall be at least 0.5mm. The main separator height over the top of the electrodes shall be at least 25mm."

3.4.6.5(f): Delete.

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3.4.11, line 4: Delete "three".

3.4.11, line 7: Delete "two layers" and substitute "any two consecutive layers of asbestos".

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3.7: Delete and substitute:

"3.7 Engineering services. Engineering services of the contractor's engineer shall be provided as specified (see 6.2.1), upon NAVSEA'S request in order to:

(a) Assist the shipyard in preparing the battery cells for installation and installing them.
(b) Assist the ship's force in becoming familiar with the battery.
(c) Take corrective action in case of unexpected problems.
(d) Examine the battery condition when requested.
(e) Participate in technical meetings with NAVSEA or the ship's force with regard to the battery and in locations other than the manufacturer's plant. Design review meetings shall not be considered a provision of engineering services.

The maximum number of mandays allowed for the entire contract or life of one set of batteries shall be as specified (see 6.2.1). Field engineering travel reports shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2)."

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3.9.1.1, line 3: Delete ", but not be limited to,".

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4.6.4.4, line 1: Delete "Upon written notification" and substitute "If notified in writing".

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5.4.1.1: Add as item (g):

"(g) Asbestos (see 3.4.6.5(b))."

6.2.1, item (h): Delete and substitute:

"(h) If engineering services are required and the maximum number of mandays allowed (see 3.7)."

PAGES 16 and 17

6.4 through 6.4.5: Delete and substitute:

"6.4 Guarantee. The guarantee of the cell when series-connect as a battery shall be as specified in 6.4.1 through 6.4.4.

"6.4.1 Nothing in the following guarantee provision shall limit any of the rights afforded to the Government by the provisions of the inspection clause including those provisions relating to latent defects, fraud, or gross mistakes that amount to fraud. This guarantee applies notwithstanding inspection and acceptance by the Government or any provision of the contract concerning the conclusiveness thereof. The contractor guarantees the performance of the battery in terms of service operating conditions (months of service or total Ah output), when the battery is operated and maintained in accordance with the service manual. The battery shall be guaranteed for 24 months or to deliver 50,000 Ah, whichever occurs first, before the battery capacity drops to 80 percent of rated capacity as specified herein, when tested at any rate in table I and subject to the temperature limitations of this specification. The period of the guarantee shall begin:

(a) Date the battery is accepted in wet condition in accordance with 4.5.2, or
(b) 24 months after acceptance in dry condition in accordance with 4.5.1, whichever occurs first.

"6.4.2 In the event of a failure of the battery or at least five cells, as specified in 6.4.3, the contractor shall:

(a) When battery capacity drops below 80 percent of rated capacity as specified herein, when tested at any rate in table I and subject to the temperature limitations of this specification within the first 12 months or 25,000 Ah, whichever occurs first, the contractor shall replace the total battery, with a new unused battery manufactured in accordance with this specification at no cost to the Government.
(b) After the period specified in (a) and prior to the end of the guarantee life of 24 months or 50,000 Ah, whichever occurs first the contractor shall indemnify the Government on a pro-rata basis (proportionately) for months of service not obtained or total Ah output not obtained, whichever is smaller. (If P being the purchase price of the battery, N the number of months not obtained, Q the total Ah output not obtained, the smaller of the two numbers $P \frac{N}{24}$ and $P \frac{Q}{50,000}$ shall be used.

"6.4.3 The battery shall be considered failed in the following cases:

(a) When grounds develop which cannot be corrected by the contractor to meet the requirements of the service manual. Corrective action by the contractor shall be at his own expense.
(b) When the battery minimum voltage specified in table I limits battery capacity to a percentage specified in 6.4.2 and cannot be restored by contractor corrective action which will be at his own expense.
(c) When five or more cells fail as defined below:

Any cell shall be considered failed in the following cases:

(a) The cell is internally shorted.
(b) The cell limits the battery capacity to a percentage specified in 6.4.2 and has to be removed in order to raise the overall battery capacity.
(c) The cell develops leaks through cracks, thus causing grounds in the system.

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"6.4.4 The same guarantee requirements of life, total Ah output, minimum capacity and indemnification specified in 6.4.1, 6.4.2 and 6.4.3 shall apply to each spare cell individually, whether wet or dry. Nothing in 6.4 diminishes the Government's rights as contained in any other provision of the contract, including but not limited to inspection."

Preparing activity:
Navy - SH
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